Pankaj Rana

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Experience

ZSW – Zentrum für Sonnenenergie- und Wasserstoff-Forschung BW

Stuttgart, Germany

Feb. 2026 – Present
Current

Research Assistant

Real-time Bat Detection on Wind Turbines · 3D Object Detection with AI

- Building Python and C++ components for the real-time LiDAR data ingestion and analysis pipeline in ROS2.
- Maintaining shared Git repositories and coordinating code contributions across a multi-developer research codebase.
- Connecting research modules into end-to-end pipelines with well-defined interfaces and clean separation of concerns.
- Writing integration and regression tests to maintain reliability across component boundaries.

May 2025 – Feb. 2026

Master's Thesis Student

Real-time Bat Detection on Wind Turbines · 3D Object Detection with AI

- Built the end-to-end object detection and tracking pipeline in ROS2, integrating LiDAR ingestion, multi-object tracking, and deep learning inference modules.
- Processed large-scale LiDAR recordings to extract temporally consistent 3D object trajectories using Kalman-based multi-object tracking.
- Developed supervised trajectory embedding models (TCN backbone with supervised contrastive learning) on synthetic motion datasets.
- Designed unsupervised autoencoder architectures for centroid-based trajectory representation learning.
- Implemented spatio-temporal BEV encodings and trained 3D U-Net models for voxel-wise occupancy prediction under extreme class imbalance.
- Evaluated models using IoU, precision, recall, F1-score, and latent space visualization (t-SNE).

Porsche Engineering Services

Nov. 2024 – Apr. 2025

Bietigheim-Bissingen, Germany

Software Development Intern

ADAS Training Simulator · Software Development & Autonomous Driving

- Developed a Windows application in C++ to control the steering wheel, seat actuator, and hardware components of the ADAS training simulator.
- Implemented real-time control loops from the CARLA simulation environment, integrated with ES Mini and OpenScenario for scenario definition.
- Worked in an agile development environment following Scrum and Kanban methodologies.

Chair of Dynamics & Control (SRS), University of Duisburg-Essen

Aug. 2024 – Feb. 2026

Duisburg, Germany

Research Assistant – CS & Programming

- Designed and developed Python programming exercises covering Data Structures and Algorithms (DSA).
- Built pytest suites to evaluate student submissions against a comprehensive set of algorithmic edge cases.

DST – Development Center for Ship Technology and Transport Systems e.V.

June 2024 – Sept. 2024

Duisburg, Germany

Research Assistant – Robotics

Cobotank · Robotics

- Implemented coordinate transformations from the AprilTag system to the robot base frame in ROS2 using C++ and Python.
- Built and optimized the 3D occupancy map (Octomap) and integrated the depth camera into the Cobotank model for environment perception.
- Debugged hardware-software interfaces and managed the project codebase with Git.

Projects

Blackboard Plotter

Oct. 2023 – Jan. 2024

University Project · UDE

Cable-driven blackboard plotter with custom 2D motion control and path-planning algorithms, implemented in embedded C++ on an ESP32 microcontroller.

Embedded C++ | ESP32

Traffic & Battery State Analysis using Machine Learning

Apr. 2024 – Oct. 2024

University Project · UDE

Classified traffic flow data using Decision Trees and Random Forests; predicted battery State of Health (SoH) and Remaining Useful Life (RUL) on the NASA Li-ion dataset via feature engineering and supervised learning.

Python | scikit-learn | Pandas | NumPy

AI Job Search Agent

Jan. 2026 – Present

github.com/pankajrana14/job-agent

Personal · Open Source

Desktop GUI automating AI-powered job search and digest delivery.

AI-powered desktop application automating job searches across LinkedIn, StepStone, and BA Jobbörse. Each posting is scored 1-10 against a free-text candidate profile via LiteLLM (GPT-4o, Claude, or Gemini with ordered model fallback). Qualified postings are deduplicated via SHA-256 hashing and delivered as HTML email digests via Gmail SMTP. Features per-scraper error isolation, multi-country search, and Windows Task Scheduler integration for twice-daily automated runs.

Python | PySide6 | Playwright | LLM Integration

Skills

LANGUAGES

English (C1) | German (B2)

PROGRAMMING LANGUAGES

C++ | Python | Embedded C++ | Bash / Shell

MACHINE LEARNING & AI

PyTorch | TensorFlow | Keras | scikit-learn | LLM Integration | Representation Learning | Autoencoders | Open3D | OpenCV | PCL | NumPy

ROBOTICS & PERCEPTION

ROS2 | LiDAR | NIR Cameras | Sensor Fusion | Point Cloud Processing | Kalman Filtering | ESP32 | CARLA | Gazebo

TOOLS & PLATFORMS

Linux | Git / GitLab | CMake | Conan | Visual Studio | Docker | GitHub Copilot | Cursor | Playwright | PySide6 · Qt | MATLAB | Simulink | Jira | pytest

Education

Apr. 2022 – Apr. 2026

Grade: 2.1 (ongoing)

Master of Science (M.Sc.)

University of Duisburg-Essen, Germany

Mechanical Engineering · Specialization in Mechatronics

Machine Learning & AI · Embedded Systems · Sensors & Actuators · Robotics & C/C++

Aug. 2016 – July 2020

Grade: 2.25

Bachelor of Engineering (B.E.)

Panjab University, India

Mechanical Engineering